

Alex Fu

afu29@bu.edu | 973-602-7076 | LinkedIn: /alex-fu-bu | Boston, MA

EDUCATION

Boston University College of Engineering, Boston, MA

May 2029

Bachelor of Science in Mechanical Engineering; Minor in Biomedical Engineering

Cumulative GPA: 3.87; Dean's List (All Semesters)

EXPERIENCE

Tymewear, Somerville, MA

June 2026 – August 2026

Engineering Intern

- Designed PCB integrating Raspberry Pi, Teensy 4.1, and stepper motor to run 100,000+ unattended stress cycles on VitalPro wearable for highly accelerated life testing and surfacing sensor output nonlinearities
- Developed Flask dashboard with headless data pipelines and live plots, enabling remote testing and monitoring for the cross-functional team
- Administered and analyzed cardiopulmonary exercise tests for 50+ athletes, translating VT1/VT2 thresholds, VO2 max, and RMV data into plain-language reports delivered post-test
- Developed and ran static-load FEA on Mark-10 force gauge mount and fabricated it in CNC-milled aluminum, reducing displacement 29-fold and eliminating fixture flex that skewed sustained-load readings
- Tuned frame-differencing CV on the rig's camera feed to auto-label test axes, removing manual annotation

PROJECTS

LymphSense, Boston, MA

September 2025 – Present

First Place Dean's Imagineering – Best Junior Team Entry

First Place BMES Design-a-thon & Student's Choice Winner

- Led mechanical integration of LymphSense, a \$150 prototype targeting the diagnostic capability of \$4,000 clinical BIS units, for stage-0 lymphedema screening at point-of-care settings at home or traveling clinics
- Modeled a mushroom-profile handheld probe housing that seats in a full-palm grip, keeping the electrode face normal to the skin without requiring fine motor control from elderly users
- Prototyped a spring-loaded 2x2 tetrapolar electrode tip with two-stage force feedback that holds contact pressure within 3-5 N, keeping electrode-skin impedance stable across readings
- Collaborated with Co-Executive Director of the National Lymphedema Network to validate clinical need, redirecting design toward point-of-care screening that gives clinicians data rather than replacing diagnosis
- Presented to biomedical engineers and industry representatives at Merck and Boston University, fielding questions on manufacturability and clinical validation pathway

SeedSwap, Boston, MA

April 2026

MIT Ripple Design-a-thon First Place Physical Track & Second Place Design for America

- Led technical design of SeedSwap, a climate-controlled seed exchange kiosk, from ideation to pitch in 48 hours, presenting renderings and system diagrams before judges from Goldman Sachs, KPMG, and Parsons
- Sized kiosk housing and interface geometry to human-factors data, using a 25° screen tilt to reduce neck strain and bin heights accommodating 95% of users
- Optimized bins visually using high-contrast colors and LED slot indicators to improve accessibility for the elderly and intuition through seed selection process

SKILLS

- **Design & Analysis:** SolidWorks (CAD, static FEA), Autodesk Fusion, Onshape, KiCAD (PCB design)
- **Software & Data:** Python, MATLAB, C++, JavaScript, Flask, OpenCV, Git, embedded systems (Teensy, Raspberry Pi), data acquisition, test automation
- **Fabrication:** CNC mill, manual lathe, OMAX waterjet, laser cutter, welding, soldering, 3D printing
- **Personal:** Swimming, Weightlifting, Trumpet

TECHNICAL PORTFOLIO

Alex Fu



B.S. in Mechanical Engineering
Minor in Biomedical Engineering
Class of 2029

My interests lie in hardware designed for human use, bridging MechE with biology and physiology.

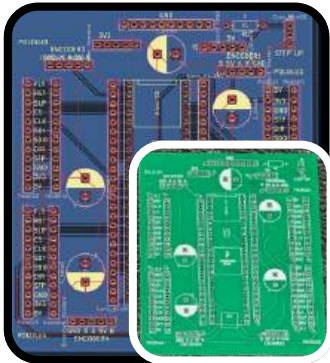


Engineering Intern

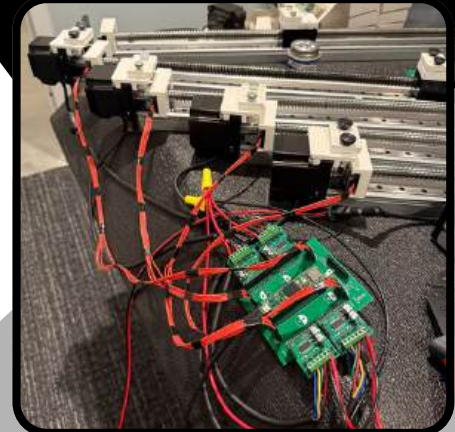
Summer 2026



Custom Circuit board

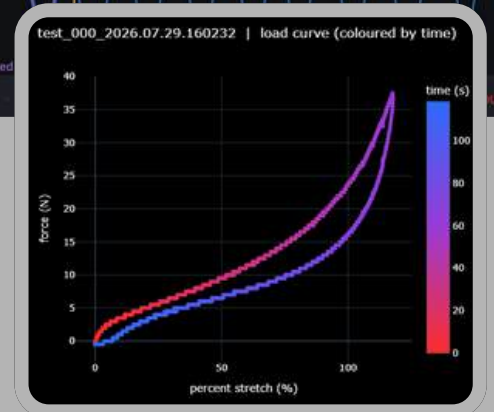


- I designed a custom PCB in KiCAD to replace the jumper wiring my system used, consolidating the whole rig onto one board so it could run without connections loosening
- It includes optimized wiring (no sharp angles, signals away from strong voltage, etc)

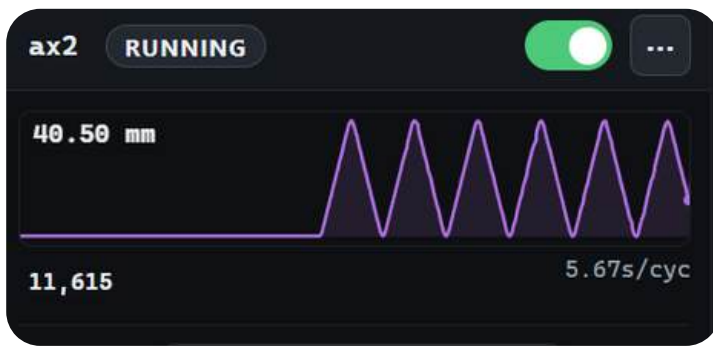


Product Testing

- The rig needed to run unattended for long durability runs, but also let someone configure and inspect a single strap in detail. I built the system on a Raspberry Pi 5 and a web interface that loads a protocol, launches it, and logs every cycle
 - **C++ (Teensy firmware):** controls motors, with microstep throttling to dramatically reduce audible noise and spinouts under fast speeds, increasing the speed cap from 50 mm/s to 250 mm/s
 - **Python + Flask (backend):** headless pipelines that run on the Pi, simultaneous motor-encoder-strap aligned recordings to compare inputs to outputs, making data analysis possible from both visual graphs and organized spreadsheet
 - **HTML, CSS, JavaScript (front end):** live position and load plots during stepped force tests and re-running past breathing data, cycle counter, and a deviation alarm that fires when the strap drifts off setpoint mid-run



- Documented the iterations and maintenance of the product testing system using AI:
 - Versioned per-module READMEs so a new session could pick up the existing architecture accurately instead of hallucinating false ones
 - Constrained changes to extending existing working code rather than rewriting them, which kept errors contained to the module I was touching
 - Automated headless data pipelines were used on the Pi rather than need repeated AI use



LymphSense

Freshman Year



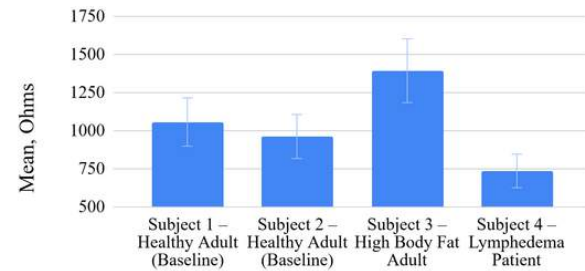
Lymphedema affect 10 million people in the USA alone. Only early stages of lymphedema are reversible, but hard to catch, many not opting to monitor high-risk areas until it's too late.

To make local lymphedema monitoring more realistically viable to incorporate into patients' lives, we've created a low-cost, portable, handheld bioimpedance spectroscopy device, LymphSense. It's designed to measure local fluid changes in areas at risk for lymphedema, poised to be used at home daily.

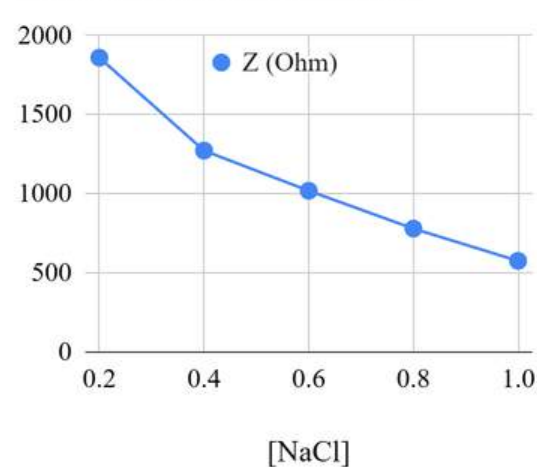
Our design won 1st place + Student Choice in the year-long BMES Design-a-thon, and Best Junior Entry for BU's Dean's Imagineering Contest.



Human Testing 50 kHz



Salt Addition vs Resistance



Repeatability Test

($p = 0.01$, 50 kHz, 400 mL water, 0.5 tsp salt)

Z (Ohm)
1269.22
969.28
1116.03
1084.65
838.35
969.38
767.65
1001.81
997.51
1039.58

Mean $\pm$ Std

1005 $\pm$ 140 Ω

What we did

- Reviewed the clinical devices (SOZO, L-Dex, LymphScanner) and DIY builds to find gap
- Interviewed Jennifer Williams, Co-Executive Director of the National Lymphedema Network. Created framing: helping doctors gather data between appointments
- Designed housing and electronics for actual manufacturing, including ideas for volume discounts and material substitutions
- Mapped the regulatory path (LymphSense is Class II, 510(k), SOZO as predicate)

Testing

- Saline: impedance dropped as salt went up, 1860 Ω to 576 Ω . Proves it tracks conductivity
- Repeatability: 1005 $\pm$ 140 Ω over 10 readings. Looser than we technically want
- Phantom: four-layer skin/fat/muscle/lymph models, one with 20% more fluid. 1075 Ω vs 784 Ω
- Human: four subjects, right leg, 50 kHz. Lymphedema at 735 Ω , healthy at 961–1055 Ω , high body fat at 1393 Ω

Phantom Testing



SeedSwap

Freshman Year

I designed SeedSwap and won first in the Ripple Design-a-thon, a 40 hour competition hosted by MIT Design for America and Wellesley In-Product.

Our team won First in the Physical Track, awarded for strong emphasis on physical usability; and Second in the Design For America (DFA) Track.

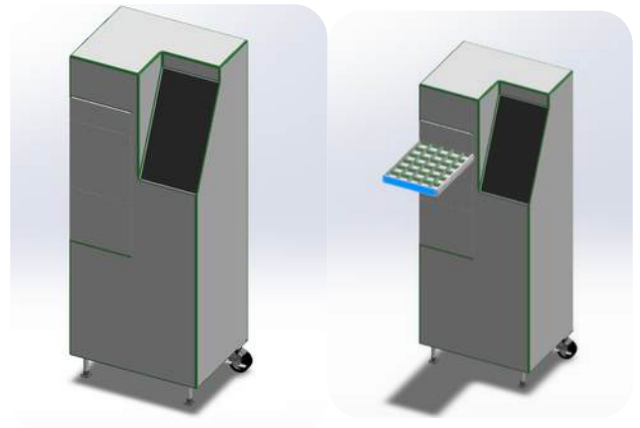
We created a vending-machine styled seed sharing device. You interact with this screen interface, optionally set up an account, and select which seeds you're inputting or taking. You don't have to swap; you can donate or just take a few.

I was the mechanical engineer in our team of three, focusing on **SolidWorks's 3D Model, Assembly, and Visualizer**

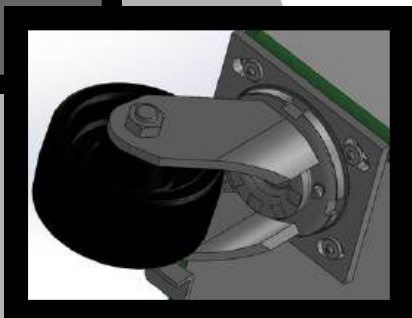
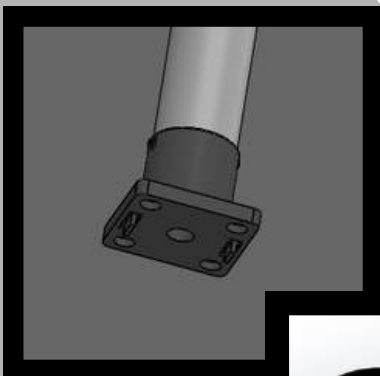
- I researched **human-centered engineering principles** to incorporate into the design. Examples
 - a 25° angled screen to reduce screen glare and neck strain
 - bin heights that would accommodate 95% of the population



Bin Demonstration



- I included mechanisms for
 - kiosk stability
 - adjustable and modular leg layout
 - visual bin cues for intuitivity + elderly accessibility
 - high contrast colors + LEDs
 - robust material selection
 - Powder-coated 16-gauge steel for longevity in outdoor environments
 - Closed-cell polyurethane foam (~1-2" thick) for seed insulation
 - climate control
 - Temperature through TMP36 + thermoelectric cooler
 - Humidity through Hygrometer Detection Module Moisture Sensor FC-28 DC 3.3V-5V IIF67124



Hybrid Feet/Wheels for Portability + Stability